4. Find the equation of the tangent to the parabola $y^2 = 16x$ perpendicular to $2x + 2y + 3 = 0$.

5. Find the equation of the tangent at $t = 2$ to the parabola $y^2 = 8x$. (Hint: use parametric form)

6. Find the equations of the tangent and normal to hyperbola $12x^2 - 9y^2 = 108$ at $\theta = \frac{\pi}{3}$. (Hint: use parametric form)

7. Prove that the point of intersection of the tangents at 't_1' and 't_2' on the parabola $y^2 = 4ax$ is $\left[at_1t_2\,,\ a\left(t_1 + t_2\right)\right]$.

8. If the normal at the point 't_1' on the parabola $y^2 = 4ax$ meets the parabola again at the point 't_2', then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

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5.7 Real life Applications of Conics

5.7.1 Parabola

The interesting applications of Parabola involve their use as reflectors and receivers of light or radio waves. For instance, cross sections of car headlights, flashlights are parabolas wherein the gadgets are formed by the paraboloid of revolution about its axis. The bulb in the headlights, flash lights is located at the focus and light from that point is reflected outward parallel to the axis of symmetry (Fig. 5.60) while Satellite dishes and field microphones used at sporting events, incoming radio waves or sound waves parallel to the axis that are reflected into the focus intensifying the same (Fig. 5.59). Similarly, in solar cooking, a parabolic mirror is mounted on a rack with a cooking pot hung in the focal area (Fig. 5.1). Incoming Sun rays parallel to the axis are reflected into the focus producing a temperature high enough for cooking.

Parabolic arches are the best stable structures also considered for their beauty to name a few, the arches on the bridge of river in Godavari, Andhra Pradesh, India, the Eiffel tower in Paris, France.



Fig. 5.49

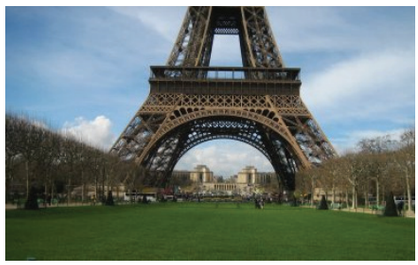
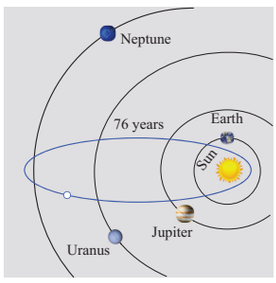


Fig. 5.50

5.7.2 Ellipse

According to Johannes Kepler, all planets in the solar system revolve around Sun in elliptic orbits with Sun at one of the foci. Some comets have elliptic orbits with Sun at one of the foci as well. E.g. Halley's Comet that is visible once every 75 years with $e \approx 0.97$ in elliptic orbit (Fig. 5.51). Our satellite moon travels around the Earth in an elliptical orbit with earth at one of its foci. Satellites of other planets also revolve around their planets in elliptical orbits as well.



The elliptical orbit of Halley's Comet

Fig. 5.51

Elliptic arches are often built for its beauty and stability. Steam boilers are believed to have greatest strength when heads are made elliptical with major and minor axes in the ratio 2:1.

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